STPM Physics · Semester 1 · Chapter 4

# WORK, ENERGY AND POWER

## 4.1 Work Done by a Force

### DEFINITIONS & LAWS

- **Work done** by a constant force is the product of the force and the displacement in the direction of the force, W = Fs cos θ, where θ is the angle between the force and the displacement; a scalar, unit joule (J).
- **One joule** is the work done when a force of 1 N moves its point of application through 1 m in the direction of the force.
- The work done by a variable force is the area under the force–displacement graph, W = ∫F dx.
- Work is positive when the force has a component in the direction of displacement, zero when the force is perpendicular to the displacement (e.g. normal reaction, centripetal force), and negative when the force opposes the displacement (e.g. friction).
- **Work–energy theorem**: the net work done on a body equals the change in its kinetic energy.
- A force is **conservative** if the work done is independent of the path (e.g. gravity, spring force); friction is non-conservative.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Work done by a constant force** | $W = Fs\cos\theta$ | Constant force; W = work done (J); F = force (N); s = displacement (m); θ = angle between force and displacement |
| **Work done against gravity** | $W = mgh$ | Raising a mass m (kg) through height h (m); g = acceleration of free fall (m $s^{-2}$); independent of the path |
| **Work done on a spring** | $W = \frac{1}{2}kx^2$ = area under the F–x graph | Spring obeying Hooke's law F = kx; work done in stretching; k = force constant (N $m^{-1}$); x = extension (m) |
| **Net work (work–energy)** | $W_{net} = \Sigma F \cdot s = \frac{1}{2}mv^2 - \frac{1}{2}mu^2$ | $W_{net}$ = net work done (J); m = mass (kg); u = initial speed (m $s^{-1}$); v = final speed (m $s^{-1}$) |
| **Work done against friction** | $W = \mu mgs$ | Level surface; μ = coefficient of friction; m = mass (kg); g = acceleration of free fall (m $s^{-2}$); s = distance moved (m) |

- 1 kW h = $3.6\times10^6$ J.

### HOW TO ANSWER (MPM STYLE)

- Write: W = Fs cos θ → substitute with the correct angle between force and displacement → answer in J with 3 s.f.
- For a variable force, state that the work done equals the area under the F–s graph, and calculate with geometric shapes (triangle, trapezium).
- For a force at an angle, use only the component of force along the displacement.
- Give the sign of work and state the energy converted (e.g. to heat for friction, to PE for lifting).

### SECTION A TRAPS

**Trap:** Using the angle between the force and the vertical (or the surface) instead of the angle between the force and the displacement; the work by weight over a vertical drop h is mgh whatever the path.

**Trap:** Zero work does not mean zero force: the normal reaction and the tension in circular motion do no work as they are perpendicular to the displacement.

**Trap:** Negative work is possible: friction and a deceleration force do negative work; the sign does not denote direction in space.

**Trap:** Area under an F–x graph below the x-axis is negative work.

### TRICKS

**Trick:** Work done by gravity depends only on vertical height difference, so ignore the path length on slopes and curves.

**Trick:** Check the unit: N m = J = kg $m^2$ $s^{-2}$; if an answer has different base units it is wrong.

**Trick:** Doubling the extension of a spring multiplies the work done by 4.

## 4.2 Conservation of Energy

### DEFINITIONS & LAWS

- **Principle of conservation of energy**: energy cannot be created or destroyed; it can only be converted from one form to another, so the total energy of an isolated system is constant.
- **Kinetic energy** is the energy a body possesses due to its motion, $E_k = \frac{1}{2}mv^2$.
- **Gravitational potential energy** is the energy a body possesses due to its position in a gravitational field; near the Earth's surface the change in potential energy is $\Delta U = mg\Delta h$.

- **Elastic potential energy** is the energy stored in a deformed spring, $U = \frac{1}{2}kx^2$.
- **Conservation of mechanical energy**: if only conservative forces do work, the sum of kinetic and potential energy is constant. If friction or air resistance acts, mechanical energy decreases and is converted to heat.
- A body released from rest from height h on a smooth surface reaches the bottom with speed $v = \sqrt{2gh}$, independent of the shape of the path.

## FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Conservation of mechanical energy** | $\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2$ | No resistive force; m = mass (kg); u, v = initial and final speeds (m $s^{-1}$); $h_1$, $h_2$ = initial and final heights (m); g = acceleration of free fall (m $s^{-2}$) |
| **Energy with friction** | $\frac{1}{2}mu^2 + mgh_1 = \frac{1}{2}mv^2 + mgh_2 + F_f d$ | Initial energy = final energy + work done against friction; $F_f$ = frictional force (N); d = distance moved against friction (m) |
| **Spring released from natural length** | $\frac{1}{2}kx^2 = \frac{1}{2}mv^2$ | Spring and mass released from the natural length; k = force constant (N $m^{-1}$); x = extension (m); m = mass (kg); v = speed (m $s^{-1}$) |
| **Spring and mass, vertical motion** | $\frac{1}{2}kx^2 + mgh$ | Energy interchange between the elastic and gravitational terms for vertical motion |
| **Simple pendulum speed** | $v = \sqrt{2gL(1 - \cos\theta)}$ | Simple pendulum released from angle θ; v = speed at the lowest point (m $s^{-1}$); L = length (m); g = acceleration of free fall (m $s^{-2}$) |
| **Loss of energy** | Loss of energy = $E_{initial} - E_{final}$ | E = energy (J) |
| **Work–energy principle** | Net work = $\Delta E_k$ | $\Delta E_k$ = change in kinetic energy (J) |

## HOW TO ANSWER (MPM STYLE)

- Identify the initial and final points, choose a zero level for potential energy, list all energy forms at both points, then equate.
- For energy lost: calculate $E_{initial} - E_{final}$ and state it is converted to heat and sound (by friction or air resistance).
- Use energy conservation when path and time are not asked, and Newton's laws when force or time is asked.
- Quote the law in words in any "state/explain" answer: total energy is constant, energy transformed from potential to kinetic, etc.

## SECTION A TRAPS

**Trap:** Total energy is always conserved; mechanical energy ($E_k$ + U) is conserved only when no friction or other non-conservative force does work. Options often mix these up.

**Trap:** Connected bodies on a string share the same speed: use $\frac{1}{2}(m_1 + m_2)v^2$.

**Trap:** Taking h as path length instead of vertical height on a slope: use the vertical height $h = s \sin \theta$.

**Trap:** At the top of a trajectory the kinetic energy is not zero when there is a horizontal velocity.

### TRICKS

**Trick:** Speed after a fall of h from rest needs no mass: $v^2 = 2gh$; mass cancels in the energy equation.

**Trick:** At the lowest point of a pendulum, all potential energy has become kinetic energy.

**Trick:** If speeds are asked at the same level, ignore the height terms altogether.

## 4.3 Power and Efficiency

### DEFINITIONS & LAWS

- **Power** is the rate of doing work (or the rate of energy transfer), $P = \frac{W}{t}$; unit watt (W), 1 W = 1 J $s^{-1}$.
- **Instantaneous power** is the power at one instant, $P = \frac{dW}{dt} = Fv$, where v is the velocity in the direction of F.
- **Average power** is the total work done divided by the total time taken.
- **Efficiency** is the ratio of the useful output energy (or power) to the total input energy (or power), usually expressed as a percentage.
- The efficiency of a real machine is less than 100% because some energy is lost as heat (friction, resistance) and sound; energy is still conserved.
- **Kilowatt-hour** is a unit of energy: the energy transferred by a power of 1 kW in 1 hour.

### FORMULAE

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Power** | $P = \frac{W}{t} = \frac{\Delta E}{t}$ | P = power (W); W = work done (J); ΔE = energy transferred (J); t = time (s) |
| **Power and force** | $P = Fv$ | F and v in the same direction; F = driving force (N); v = velocity (m $s^{-1}$) |

| Quantity / law | Formula | Symbols, units & conditions |
|---|---|---|
| **Power to raise a load** | $P = mgv$; $P = \frac{mgh}{t}$ | Raising a load at constant speed; m = mass (kg); g = acceleration of free fall (m $s^{-2}$); v = speed (m $s^{-1}$); h = height (m); t = time (s) |
| **Efficiency** | η = (useful energy output/energy input) × 100% = (useful power output/power input) × 100% | η = efficiency (%); energies in J; powers in W |
| **Power balance** | Input power = output power + power wasted | Powers in W |
| **Vehicle on a slope** | F = resistive force + mg sin θ; P = Fv | Vehicle at constant speed on a slope of angle θ; F = driving force (N); m = mass (kg); g = acceleration of free fall (m $s^{-2}$); v = speed (m $s^{-1}$) |
| **Power of flowing fluid** | $P = \frac{1}{2}\frac{dm}{dt}v^2 = \frac{1}{2}\rho A v^3$ | Flowing fluid (e.g. water jet or wind); $\frac{dm}{dt}$ = mass flow rate (kg $s^{-1}$); v = speed (m $s^{-1}$); ρ = density (kg $m^{-3}$); A = cross-sectional area ($m^2$) |

- 1 kW h = $3.6\times10^6$ J.

## HOW TO ANSWER (MPM STYLE)

- State the relation used, substitute with units in SI (convert kW to W, minutes to s), and give the answer with unit and 3 s.f.
- For efficiency problems, identify the useful output and the input: write $\eta = \frac{P_{out}}{P_{in}}$ then rearrange.
- For a vehicle at maximum speed, state that the acceleration is zero so the driving force equals the total resistive force, then P = Fv.
- For a heated or lifted mass, find the useful energy first (mgh, mcΔθ) then divide by time.

## SECTION A TRAPS

**Trap:** Mixing the efficiency ratio and percentage: η = 0.80 is 80%; a result above 100% is impossible.

**Trap:** Using P = Fv with the wrong F: at constant speed F balances resistance (plus weight component on a slope), it is not the weight alone.

**Trap:** For the same height and time, power is directly proportional to the mass lifted ($P = \frac{mgh}{t}$); doubling the mass doubles the power.

**Trap:** Energy and power confusion: kW h is energy, kW is power; a larger power does not mean more work unless the time is considered.

## TRICKS

**Trick:** Doubling the speed against a constant resistive force doubles P ($P = Fv$); if resistance $\propto v^2$ then $P \propto v^3$.

**Trick:** For efficiency chains (motor then pump), the overall efficiency is the product of the individual efficiencies.

**Trick:** Check units: $W = J\ s^{-1} = N\ m\ s^{-1} = kg\ m^2\ s^{-3}$.